



Healthy diet in adults

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INTRODUCTION

Poor diet is one of the leading causes of mortality in the United States, along with tobacco use, hypertension, and overweight, and is a leading cause for disability-adjusted life years [1].

Decades of research have revealed strong links between many dietary components and eating patterns and important health outcomes [2]. Furthermore, the majority of patients and clinicians view eating habits as important to health and an important part of healthy lifestyle counseling [3,4].

This topic will discuss the evidence on the health effects of various types of dietary patterns, dietary components, and select foods. Although the science on the larger environmental effects of food choices, and their potential impact on health, continues to advance, it is an issue that is outside of the current scope of this review [5].

Dietary assessment and diets for specific populations are discussed elsewhere:

- (See "[Dietary assessment in adults](#)".)
 - (See "[Dietary history and recommended dietary intake in children](#)".)
 - (See "[Obesity in adults: Dietary therapy](#)".)
 - (See "[Nutrition in pregnancy: Assessment and counseling](#)".)
 - (See "[Nutrition in pregnancy: Dietary requirements and supplements](#)".)
 - (See "[Nutritional issues in older adults](#)".)
 - (See "[Medical nutrition therapy for type 2 diabetes mellitus](#)".)
 - (See "[Lipid management with diet or dietary supplements](#)".)
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IMPACT OF A HEALTHY DIET

While the specifics of healthy eating can vary, the overall goal of a healthy diet is to improve patients' health and lower the risk of disease. Considerable evidence shows us that healthy food choices can extend longevity [6] and reduce the risk of cardiovascular disease, diabetes, hypertension, and cancer, among other important chronic

diseases. A healthy diet can also help with weight management and is associated with improved health-related quality of life and a decreased risk of frailty in older adults [7-11].

With such a broad impact, population-wide improvements in diet could have a substantial impact not only on population health but also on health care costs. Considering only cardiometabolic disease, a 2019 analysis using the United States National Health and Nutrition Examination Survey (NHANES) estimated the total annual costs related to poor diet to be approximately USD \$50 billion [12]. Obesity-related medical costs have been estimated at \$150 billion USD per year [13].

CHOOSING A HEALTHY DIET: PATIENT COUNSELING

Choosing a dietary pattern — The specific diet or dietary pattern an individual chooses is strongly influenced by a wide range of factors, including personal preference, social and cultural influences, as well as economic considerations. While maintenance of good health or prevention of disease may be important drivers of dietary choices, these will generally not be the patient's sole considerations; taste preferences, cost and food availability issues, religious requirements, social norms, and media and marketing forces all contribute. As an example, the lower cost of plant-based diets or their impact on planetary health (ie, reducing adverse climate change, promoting environmental sustainability) may be important considerations for some patients [14,15].

Clinicians should work with individuals to determine the best healthy diet pattern based upon their unique circumstances. Patient encounters provide a valuable opportunity to promote healthy food choices. In a 2017 systematic review evaluating the impact of behavioral interventions, there were multiple, though small, benefits from behavioral interventions on diet and physical activity, serum lipids, body mass index, and blood pressure at 6 to 12 months [16]. While rates of such counseling have been increasing in the United States, however, in 2016, only around one-third of patients reported receiving dietary advice from clinicians or other medical professionals [17].

Maintaining energy balance — Although weight management is not the focus of the diets in this review, any discussion of healthy diet needs to emphasize the importance of maintaining energy balance over time. Overnutrition leading to overweight and obesity is the single most important dietary factor associated with poor health outcomes. It is associated with premature mortality as well as increased incidence of cardiovascular disease, diabetes, hypertension, cancer, and other important conditions [18-20].

Whatever dietary pattern an individual follows, maintenance of energy balance is an important overall goal. Avoidance of weight gain, even if a patient is overweight or has obesity, can have important long-term health benefits.

Total energy expenditure for recommended daily caloric intake is based upon age, sex, weight, and activity level ([table 1](#)) (see "[Obesity: Genetic contribution and pathophysiology](#)", section on '[Physiologic processes affecting energy balance](#)'). Calculating a patient's daily caloric intake can be aided by using 24-hour dietary recall, a food diary, or other assessment tools to determine intake of different foods and beverages. Assessment tools, including consumer-focused apps and online tools that can help patients readily track what they eat and drink, are discussed separately. (See "[Dietary assessment in adults](#)".)

SPECIFIC HEALTHY DIETARY PATTERNS

Dietary guidelines: Recommended dietary patterns — Specific dietary guidelines may vary, but in general, a high-quality, healthy diet emphasizes limited consumption of red and processed meats, unhealthy fats (saturated and industrial trans fats), sugar, sodium, and alcohol, and a greater intake of fruits, vegetables, legumes, nuts, and whole grains. This pattern is consistent with that generally recommended by the World Health Organization

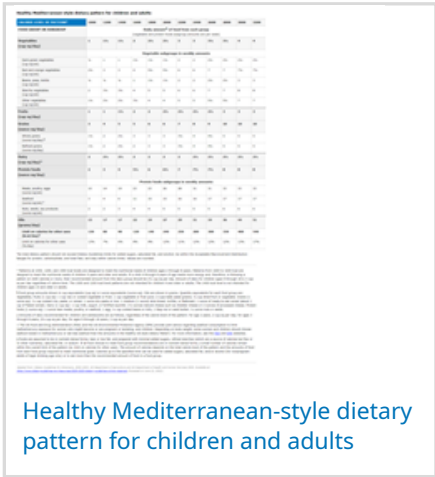
[21], the World Cancer Research Fund/American Institute for Cancer Research [22], the American Cancer Society [23], the American Heart association [24], and the US Department of Agriculture [25,26], as well as other groups [27]. In addition to multiple disease-specific benefits, results of meta-analyses and individual observational studies have found consistent associations between this dietary pattern and decreased mortality [28-35]. This dietary pattern has also been associated with an increased likelihood of healthy aging (ie, aging without impairments in cognitive, physical, or mental health) [36].

The development of healthy eating guidelines can vary substantially, from expert panel review of existing evidence to the development and analysis of guideline-specific data [22]. While differences in methods may impact the detailed conclusions of guidelines as well as the confidence in them, there is generally consistency in messaging, both among organizations and even countries [25,37-39].

The evidence evaluating specific dietary patterns can vary substantially, and it is likely that some diets that patients inquire about have been little studied. Although long-term randomized controlled trials are often absent, observational studies support the association of health benefits with a number of dietary patterns.

Most of the dietary patterns discussed below share some similar aspects but are further defined by regional characteristics or a specific focus or restriction on dietary components or types of foods.

Mediterranean diet — There is no single definition, but a Mediterranean diet is typically high in fruits, vegetables, whole grains, beans, nuts, and seeds. A Mediterranean diet includes olive oil as an important source of monounsaturated fat and allows low to moderate wine consumption. It generally includes low to moderate amounts of fish, poultry, and dairy products, and it includes little red meat ([table 2](#)). Although the Mediterranean diet is associated with several health benefits, it remains uncertain if a single component of the Mediterranean diet is associated with the protective benefit or if the benefit results from an aggregation of effects.



Healthy Mediterranean-style dietary pattern for children and adults

Table 2 - larger image below

- In a meta-analysis of randomized trials including the large PREDIMED trial [40], a Mediterranean diet reduced the risk of stroke compared with a low-fat diet (hazard ratio [HR] 0.60, 95% CI 0.45-0.80) but did not reduce the incidence of cardiovascular or overall mortality [41].
- In a subsequent 2022 trial including over 1000 adults with coronary heart disease conducted in Spain (CORDIOPREV), randomization to a Mediterranean diet resulted in fewer major cardiovascular events (myocardial infarction, revascularization, ischemic stroke, peripheral artery disease, cardiovascular death) compared with a low-fat diet at seven years' follow-up [42].
- In a network meta-analysis of 40 trials comparing seven dietary patterns among 35, 548 participants at increased risk of CVD, a Mediterranean diet (12 trials) resulted in lower risk of several outcomes including all-cause mortality (odds ratio [OR] 0.72, 95% CI 0.56-0.92), cardiovascular mortality (OR 0.55, 95% CI 0.39-0.78), stroke (OR 0.65, 95% CI 0.46-0.93), and nonfatal myocardial infarction (OR 0.48, 95% CI 0.36-0.65), when compared with minimal intervention [43].
- In observational studies, a Mediterranean diet has been associated with lower overall mortality and cardiovascular mortality [44-48], as well as a decreased incidence of Parkinson disease, Alzheimer disease, and cancers, including colorectal, prostate, aerodigestive, oropharyngeal, and breast cancers [45,46,49-51]. (See "Factors that modify breast cancer risk in females", section on 'Diet rich in fruits and vegetables, fish, and olive oil (eg, Mediterranean diet)').

The effects of the Mediterranean diet on type 2 diabetes mellitus and macular degeneration are discussed separately. (See ["Medical nutrition therapy for type 2 diabetes mellitus"](#), section on 'Macronutrient composition' and ["Age-related macular degeneration"](#), section on 'Diet/alcohol'.)

DASH diet — The [DASH \(Dietary Approaches to Stop Hypertension\) diet](#) is composed of four to five servings of fruit, four to five servings of vegetables, and two to three servings of low-fat dairy per day, with <25 percent of daily caloric intake from fat.

The DASH diet has been studied in both normotensive and hypertensive populations and has been found to lower systolic and diastolic pressure more than a diet rich in fruits and vegetables alone [52]. The combination of low-sodium and DASH diet resulted in further decreases in blood pressure, comparable with those observed with antihypertensive agents [53]. (See ["Salt intake and hypertension"](#) and ["Diet in the treatment and prevention of hypertension"](#).)

The DASH diet has also been associated with a lower risk of colorectal cancer, cardiovascular disease, premature mortality, and gout (in males) [34,54-58]. These benefits have not been confirmed in randomized trials [59], although available trials all have fairly short follow-up periods (up to 18 months), limiting conclusions about the impact on chronic diseases. (See ["Gout: Nonpharmacologic strategies for prevention and treatment"](#), section on 'Modifying diet'.)

Plant-based and vegetarian diets — Plant-based diets, which include vegetarian (and similar) diets, focus on the consumption of cereals, fruits, vegetables, legumes, and nuts, with the restriction of animal-based foods, including meat, dairy products, and eggs ([table 3](#)) [60]. The degree of animal food restriction in plant-based eating varies considerably [61]. Vegan diets exclude all animal-based foods, while less restrictive approaches can include eggs, dairy products, and some meat. Selected plant-based diets may be grouped as follows:

- **Vegan** – All animal products, including eggs, milk, and milk products, are excluded from the diet. Some vegans do not use honey and may refrain from using animal products such as leather or wool. They also may avoid foods that are processed or not organically grown [62].
- **Lactovegetarian** – Milk and milk products are included in the diet, but no eggs or meat are consumed.
- **Lacto-ovovegetarian** – Eggs, milk, and milk products (lacto = dairy; ovo = eggs) are included, but no meat is consumed.
- **Semi-vegetarian (flexitarian)** – Meat occasionally is included in the diet. Some people who follow such a diet may not eat red meat but may eat fish (pescatarian) or chicken (pollotarian). The Mediterranean diet could be considered a semi-vegetarian, plant-based diet. (See ["Mediterranean diet"](#) above.)
- **Macrobiotic** – Vegetables, fruits, legumes, and seaweeds are included in the diet, while whole grains, especially brown rice, are also emphasized. Locally grown fruits are recommended. Animal foods are generally limited to white meat or fish, which may be included in the diet once or twice a week.

The long-term effects of a plant-based diets on health outcomes may be difficult to separate from lifestyle factors associated with consumption of a plant-based diet (eg, regular exercise, avoidance of tobacco and alcohol products). However, observational studies suggest that consumption of plant-based and vegetarian diets is associated with lower risk of obesity, coronary heart disease, hypertension, type 2 diabetes, certain cancers, and possibly all-cause mortality [63-74]. Results of randomized trials have found a beneficial impact of vegetarian diets on a number of cardiovascular risk factors [75,76]. As an example, in a meta-analysis of 20 randomized trials (n = 1878) of individuals with or at high risk for cardiovascular disease, following a vegetarian diet for an average of six months reduced the cardiovascular risk factors of LDL-C (low density lipoprotein cholesterol), hemoglobin A1C, and weight by 6.6 mg/dL, 0.24 percent, and 3.4 kg, respectively [77]. Although

one observational study suggested a modestly increased risk of stroke with a vegetarian diet in older adults (at approximately 85 years old), the risk estimate was imprecise and has not been observed in other studies [73].

The nutritional adequacy and quality of plant-based and vegetarian diets should be judged individually, based on the type, amount, and variety of nutrients that are consumed [78]. As an example, some but not all studies suggest that vegan diets (which exclude all animal-based products, including dairy) may be associated with lower bone mineral density and higher fracture risk due to lower intake of calcium [79,80]. In addition, individuals whose diet excludes all animal products (ie, meat, eggs, milk, and milk products) are at risk for inadequate [vitamin B12](#) intake and may benefit from vitamin B12 supplementation. (See "[Vitamin intake and disease prevention](#)", [section on 'Special diets'](#).)

Additionally, as with other healthy eating patterns, plant-based diets should emphasize the consumption of lesser amounts of processed foods as well as foods that are low in unhealthy fats, refined grains, sugar, and sodium.

Low-fat diet — Low-fat diets focus on limiting calories from fat. Some approaches are very low-fat, with calories from fat ≤ 10 percent of total calories; others are more moderate, with ≤ 30 percent of daily caloric intake from fat.

Most evidence also points to little or no health or mortality benefit from a low-fat dietary pattern [81-83], although other analyses suggest some possible mortality, coronary heart disease, diabetes, and breast cancer survival benefits [43,84].

Compared with higher-fat diets, lower-fat diets are associated with lower body adiposity [85]. Overall, however, low-fat diets have been found to be no more effective than other types of diets for weight control [86,87]. This is discussed in detail separately. (See "[Obesity in adults: Dietary therapy](#)", [section on 'Low-fat diets'](#).)

Individuals choosing a reduced-fat diet should focus on food quality in addition to fat content. A 2020 prospective cohort study using data from the US National Health and Nutrition Examination Survey (NHANES) found that an unhealthy low-fat diet was associated with a modest increase in total mortality, while a healthy low-fat diet was associated with a modest decrease in total mortality [88]. While consuming a reduced-fat diet, the emphasis should be on whole grains, legumes, fruits, and vegetables, with limited refined grains and sugar; fat should be from healthy sources, such as fatty fish, olive oil, and canola oil.

Low-cholesterol diet — A low-cholesterol diet focuses on limiting the amount of dietary cholesterol, which can be found in higher levels in foods such as certain shellfish, full-fat dairy, and eggs.

While elevated serum levels of low-density lipoprotein (LDL) cholesterol are an established risk factor for the development of cardiovascular disease, the relationship between dietary cholesterol intake and cardiovascular disease is less certain [89,90]. Because of their cholesterol content, eggs have often been restricted in low-cholesterol diets. A 2020 systematic review and meta-analysis of egg consumption found no association between daily intake of a single egg and risk of cardiovascular disease [91]. Similar overall null results were found for type 2 diabetes, although a small increase in risk with moderate egg consumption was seen in United States study populations [92]. However, in a subsequent 2021 observational study including over 96,000 postmenopausal females without known cardiovascular disease at baseline, higher egg consumption (≥ 1 egg/day versus < 1 egg/week) was associated with greater risk of incident cardiovascular disease and total mortality (HR 1.14, 95% CI 1.04-1.25; and HR 1.14, 95% CI 1.07-1.22, respectively) [93].

For cardiovascular disease risk reduction, a diet low in saturated fat is the preferred approach. (See "[Unhealthy fats](#)" below.)

DIETARY COMPONENTS ASSOCIATED WITH HEALTH BENEFITS

Although we focus on overall dietary patterns, individual dietary components can play an important role in nutrition and health outcomes and can offer patients specific areas to focus their efforts at improving their diet.

Fruits and vegetables — Patients should be counseled to consume three servings of vegetables and two servings of fruit per day for a 2000 calorie diet ([table 4](#)) [25]. Fruits and vegetables are a rich source of fiber, essential vitamins and minerals, and other potentially beneficial compounds, as well as carbohydrates with a low glycemic index. (See 'Fiber' below and "[Vitamin intake and disease prevention](#)".)

In large, prospective cohort studies, increased consumption of fruits and vegetables is associated with decreased all-cause and coronary heart disease mortality [94-97]. Several studies suggest that fruit and vegetable consumption is associated with a lower risk of coronary heart disease and stroke [98-100]. While large prospective cohort studies have not found a substantial effect of fruits and vegetables on cancer incidence in the general population, there is evidence of impact on some specific cancers, including aerodigestive cancers [101] and estrogen receptor-negative breast cancer. Some observational studies also support an association between higher intake of tomatoes and tomato-based products with a lower risk of prostate cancer [102,103]. (See "[Overview of cancer prevention](#)", [section on 'Fruits and vegetables'](#)".)

Whole grains — Individuals should consume at least two to four servings a day of whole grains, which tend to be fiber-rich, as recommended by the Dietary Guidelines for Americans ([table 5](#)) [25,26].

Common types of food made from grain include breads, breakfast cereals, oatmeal, tortillas, and pasta. Grain products vary by level of milling or processing and can be defined as follows:

- Refined grains (eg, white rice, white bread, refined and sweetened cereals) have bran and germ removed during processing, which also removes fiber, iron, B vitamins, and other nutrients. Processing produces a finer texture and improves shelf life.
- Enriched grains are refined grains that have iron and B vitamins added back, but not fiber in most cases.
- Whole-grain foods include brown rice, whole-wheat bread, whole-grain cereal, and oatmeal. These are a good source of fiber and other nutrients and are considered carbohydrates with a lower glycemic index.

Increased consumption of whole grains is associated with a variety of health benefits, including improved weight management, reduction in cardiovascular and all-cause mortality, and decreased incidence of coronary heart disease and cancer [104-107]. Representative studies include:

- In a prospective study of over 120,000 healthy individuals, an increased intake in refined grains was associated with long-term weight gain, whereas an increased intake in whole grains was associated with weight loss [105].
- In a 2014 meta-analysis of 18 studies including over 400,000 participants, increased whole-grain intake was associated with decreased risk for coronary heart disease (relative risk [RR] 0.79, 95% CI 0.74-0.83) [106].
- In a pooled prospective cohort study including over 117,000 individuals, higher whole-grain consumption was associated with lower cardiovascular (RR 0.85, 95% CI 0.78-0.92) and total (RR 0.91, 95% CI 0.88-0.95) mortality

USDA recommended daily or weekly caloric intake of different food groups (USDA recommended average daily food intake)

Food Group	Calories per Day	Calories per Week
Grains	480	3,360
Vegetables	250	1,750
Fruits	150	1,050
Protein Foods	150	1,050
Dairy	150	1,050
Oils	50	350
Alcohol	0	0
Total	1,030	7,210

Table 5 - larger image below

[107].

The World Cancer Research Fund/American Institute for Cancer Research, in their Third Expert Report: Diet, Nutrition, Physical Activity and Cancer, recommends increased dietary intake of whole grains to reduce the risk of developing colorectal cancer [101].

Fiber — The recommended amount of dietary fiber is approximately 14 g per 1000 calories, which translates to approximately 25 g to 34 g per day for most adults ([table 6](#)) [25].

Fiber is the portion of plants that cannot be digested by enzymes in the gastrointestinal tract. Fiber is available in a large variety of natural foods and supplements ([table 7](#)). Patients should be advised to replace refined grains (eg, white rice, white bread) with whole grains (eg, brown rice, whole-wheat bread), which have higher fiber content.

Increased fiber intake is associated with many health benefits, including a decreased risk of coronary heart disease, cardiovascular and all-cause mortality, colorectal cancer, stroke, and type 2 diabetes [101,104,108-116]. High-fiber diets may in part protect against coronary heart disease by controlling cardiovascular risk factors, including lowering insulin levels and blood pressure, and improving lipid profiles [117,118].

The benefit of increased fiber intake is also observed in people with established cardiovascular disease. For example, in a prospective cohort study following 4000 patients after a first myocardial infarction, eating a high-fiber diet (post-myocardial infarction) was associated with lower cardiovascular and all-cause mortality (RR 0.72, 95% CI 0.52-0.99; and RR 0.73, 95% 0.58-0.91, respectively) [119]. The largest benefit was seen among those with the greatest increase in fiber consumption, comparing pre- and post-myocardial infarction fiber intake.

Further details on dietary fiber in managing and preventing disease are discussed separately. (See "[Nutritional considerations in type 1 diabetes mellitus](#)" and "[Medical nutrition therapy for type 2 diabetes mellitus](#)" and "[Overview of cancer prevention](#)", section on 'Fruits and vegetables' and "[Type 2 diabetes mellitus: Prevalence and risk factors](#)", section on 'Dietary patterns' and "[Epidemiology and risk factors for colorectal cancer](#)", section on 'Fiber'.)

Reduced-fat dairy — The United States Dietary Guidelines for Americans recommends that adults consume three servings of dairy products daily for a 2000-calorie diet ([table 5](#)) [25].

The dairy food group is composed of milk and foods made from milk, such as yogurt and cheese. Dairy products are a good source of protein, calcium, vitamin D, and potassium. Diets that include limited amounts of dairy should ensure adequate intake of these nutrients from other sources.

Intake of dairy products likely lowers the risk of colorectal cancer [101] and possibly cardiovascular disease [120-122]. Low-fat dairy is also integral part of the DASH diet, which has been shown to help control blood pressure as well as lower the risk of colorectal cancer, cardiovascular disease, and premature mortality. (See '[DASH diet](#)' above.).

The data regarding the cardiovascular effects of dairy are conflicting [123]. Some observational studies have found an association between dairy consumption and lower risk of cardiovascular disease and mortality. As an example, in a multinational cohort study including over 130,000 patients followed for more than nine years, higher intake of total dairy was associated with lower total mortality, cardiovascular mortality, major cardiovascular disease, and stroke [122]. However, other studies have not found such an association. Moreover, it is uncertain whether consuming low-fat versus full-fat dairy has differential health effects. Most guidelines have traditionally recommended low-fat dairy, largely due to concerns around saturated fat content of full-fat dairy. However, meta-analyses of observational studies have not consistently found a differential health effect of low-fat versus full-fat dairy [124,125].

Protein-rich foods — We agree with the 2020 to 2025 Dietary Guidelines for Americans, which recommend that individuals consume 5.5 ounces of protein-rich foods daily for a 2000 calorie diet ([table 5](#)) [25]. This is equivalent to approximately 35 to 45 grams of protein daily, or approximately 0.5 to 0.6 grams per kilogram for a 70-kilogram adult, which is the minimum needed to maintain a neutral nitrogen balance [126]. Healthy protein-rich foods include poultry, beans, peas, seeds, nuts, fish, and soy products (eg, tofu and tempeh). (See 'Healthy fats' below.)

In contrast, the 2025 to 2030 Dietary Guidelines for Americans recommend a higher protein intake goal of 1.2 to 1.6 grams of protein per kilogram of body weight per day (approximately 80 to 112 grams for a 70 kilogram adult) [26]. However, there is no high-quality evidence to support this higher goal in the general population.

Nuts — Nut consumption is associated with lower risk for cardiovascular disease. One study that combined data from three large cohorts found that the risk of myocardial infarction or stroke was reduced among participants who consumed nuts five or more times per week compared with participants who rarely ate nuts (hazard ratio [HR] 0.86, 95% CI 0.79-0.93) [127]. Peanuts, tree nuts, and walnuts were all associated with similar reductions in the risk of cardiovascular disease.

Fish — One to two servings (total of eight ounces) of oily fish (eg, salmon and tuna) per week is suggested for most adults, with special consideration regarding mercury content of particular seafoods for individuals who are pregnant, planning pregnancy, and breastfeeding [25,128]. Reference levels for mercury in commercial fish and shellfish are available [online](#), and information for patients is available from the US Food and Drug Administration (FDA) ([list of fish categorized by mercury concentration](#)). (See "Fish consumption and marine omega-3 fatty acid supplementation in pregnancy", section on 'Fish'.)

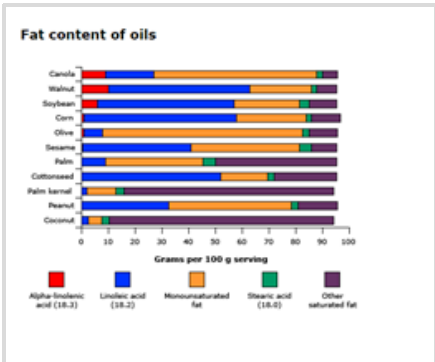
In a 2014 meta-analysis of 11 prospective cohort studies, fish consumption ≥ 4 times per week was associated with a decreased risk of acute coronary syndrome (RR 0.79, 95% CI 0.70-0.89) with a dose-response relationship (each additional 100 g serving per week associated with RR 0.95, 95% CI 0.92-0.97) [129]. In a subsequent pooled analysis of several international cohort studies, consuming two servings of fish weekly was associated with lower risk of major CVD and mortality among persons with existing CVD (HR 0.84, 95% CI 0.73-0.96; and HR 0.82, 95% CI 0.74-0.91, respectively), but not in those without CVD [130]. (See "Overview of primary prevention of cardiovascular disease in adults", section on 'Omega-3 fatty acids'.)

Healthy fats — Saturated fat should make up no more than 10 percent of total caloric intake, as recommended by the Dietary Guidelines for Americans [25].

The type of fat consumed appears to be more important than the amount of total fat ([table 8](#)). Individuals should choose poly- and monounsaturated fats in place of saturated and trans fats [131] (see 'Unhealthy fats' below). Consumption of polyunsaturated fats can improve serum lipids and lower the risk of CVD and total mortality [131-134]. In a meta-analysis of four randomized controlled trials including individuals without coronary heart disease, replacement of dietary saturated fat with polyunsaturated fat reduced the risk of coronary heart disease by 29 percent compared with no intervention (RR 0.71, 95% CI 0.62-0.81) [131]. (See "Dietary fat", section on 'Type of fat' and "Lipid management with diet or dietary supplements".)

Good sources of polyunsaturated and monounsaturated fats are certain oils (eg, olive, canola, and peanut) ([figure 1](#)), as well as avocados, certain nuts, and fatty fish.

DIETARY PATTERNS WITH LIMITED OR DEVELOPING EVIDENCE



The list of diets with limited evidence of benefit is very long and potentially unlimited, as new diets and eating patterns regularly emerge both in the medical literature and popular media. Here, we highlight diets that patients may be likely to inquire about or that show potential evidence of health benefits.

Figure 1 - larger image below

Low-carbohydrate diet — Low-carbohydrate diets emphasize protein and fat intake with limited carbohydrate intake. The level of carbohydrate restriction in low-carbohydrate diets can vary significantly.

Ketogenic diets (high in dietary fat and low in carbohydrates, which causes the body to produce ketones from body fat to use for fuel) can have very few carbohydrates, often under 50 g per day, while some low-carbohydrate diets can have 40 percent of total calories from carbohydrates. The more restrictive the diet, however, the more difficult long-term maintenance can be for patients.

Low-carbohydrate diets may aid in weight loss and possibly improve serum lipids over the short term, although results for low-density lipoprotein (LDL) cholesterol are mixed [135,136]. In addition, the health effects of low-carbohydrate diets over the longer term remain unclear and may depend upon overall dietary food quality; unhealthy low-carbohydrate diets (eg, higher in animal protein and saturated fat) have been found in observational studies to be associated with an increased mortality risk, while healthy low-carbohydrate diets (higher in plant protein and unsaturated fats) associated with a lower risk [88,137,138].

Organic diet — There is significant interest in the potential health benefits of eating an organic diet. Although there are regulatory definitions for what qualifies as organic produce and organic egg, meat, and dairy products [139], there is a lack of established methodology in assessing what actually qualifies as an organic diet. In addition, complicating the research on the impact of organic food on health is the difficulty of taking fully into account the potential confounders linked with organic food consumption (eg, other elements of a healthy lifestyle) [140].

Despite great interest in the area, high-quality data are still lacking on the links between the consumption of organic food and improved health. While eating a diet high in organic foods can lessen exposure to synthetic pesticides, it is unclear if this actually translates to improved health outcomes [140,141]. Growing evidence does, however, point to a possible link between high organic food intake and a lower risk of cancer, particularly non-Hodgkin lymphoma [142,143]. Further evidence from well-designed studies is needed to confirm this association.

Anti-inflammatory diet — The anti-inflammatory diet focuses on food choices that may reduce systemic inflammation. Chronic inflammation is linked to an increased risk of multiple chronic diseases, including cardiovascular disease, type 2 diabetes, and cancer [144].

There is no specific anti-inflammatory diet, and aspects of other healthy diets qualify as anti-inflammatory (see '[Mediterranean diet](#)' above). Overall, however, an anti-inflammatory diet emphasizes choices such as olive oil, nuts, fruits, leafy green vegetables, and fatty fish; and limits processed meats, refined grains, sugar, and fried food. Compared with anti-inflammatory diets, those diets that "promote" inflammation have been linked in observational studies with multiple health risks, including an increased risk of total, cancer, and cardiovascular disease-related mortality [144-146]. The science on anti-inflammatory diet continues to develop.

Gluten-free diet — A gluten-free diet eliminates or substantially limits grains that contain the protein gluten, such as wheat and barley ([table 9](#)). In individuals with celiac disease and non-celiac gluten sensitivity, a gluten-free diet has demonstrated benefits (see "[Management of celiac disease in adults](#)" and "[Celiac disease in children and adolescents: Management](#)"). Outside of those with gluten sensitivity, however, little evidence supports health benefits from a gluten-free diet, and some evidence suggests increased risk of cardiovascular disease due to related reductions in whole grain intake [147].

DIETARY PATTERNS AND COMPONENTS TO LIMIT OR AVOID

There is good evidence associating certain dietary patterns and individual dietary components with adverse health outcomes. Individuals should be encouraged to limit and, in some instances, avoid these components. Realistically, however, elimination may be too drastic for many patients, and a reduction in consumption is a more attainable goal, at least initially.

"Western" diet — Studies consistently show that the typical Western diet increases the risk of multiple chronic diseases. The Western diet is marked by consumption of red and processed meats, full-fat dairy products, refined grains, high-sodium foods, and sugar, as well as low amounts of fiber, whole grains, legumes, and fruits and vegetables.

Compared with healthy eating patterns, the Western diet is linked to an increased risk of cardiovascular disease, type 2 diabetes, cancer, overweight and obesity, and other conditions. In 2016, poor diet was the leading cause of death and the third leading cause for disability-adjusted life-years in the United States [148]. Individuals should be encouraged to avoid the typical Western diet in favor of healthier eating patterns that are more plant-based, with a focus on healthy fats and fewer processed products.

Unhealthy fats — Individuals should limit saturated and trans fats, replacing them with healthy fats such as polyunsaturated and monounsaturated fats. (See 'Healthy fats' above and "Dietary fat", section on 'Type of fat'.)

- **Saturated fat** – Saturated fat should make up no more than 10 percent of total caloric intake, as recommended by the Dietary Guidelines for Americans [26], as it has been shown to adversely impact serum lipids and increase the risk of cardiovascular disease [131]. As an example, in a meta-analysis of four randomized trials with a minimum two-year duration, replacing saturated fat with polyunsaturated fat reduced the risk of coronary heart disease by almost 30 percent (RR 0.71, 95% CI 0.62 to 0.81) [131].

The effect of reducing saturated fat on mortality is less certain. In a meta-analysis of randomized trials, lower saturated fat intake reduced cardiovascular mortality (RR 0.93, CI 0.77 to 1.11) and all-cause mortality (RR 0.96, 95% CI 0.88-1.06), but the confidence intervals were wide and included no effect [149]. The absolute benefit was greatest in those with higher baseline cardiovascular risk and most pronounced when replacing saturated with polyunsaturated fat. However, most of the included studies were at high risk of bias due to methodologic limitations (eg, lack of blinding).

Saturated fats are common in animal products, such as red meat and whole-fat dairy, and in some plant products, such as coconut and palm oils.

- **Trans fats** – Trans fats used to be common in many foods in the United States but since 2018 have been largely removed from the food supply following a US Food and Drug Administration ban on their use [150]. However, some foods can still include some trans fat, such as some fast foods and shortenings and oils with hydrogenated oil listed in the ingredients.

Added sugars and sugar-sweetened beverages — Added sugars should be avoided in the diet [25,26]. These sugars often come from sweetened beverages and processed foods. If a higher percentage of caloric intake comes from added sugars, it may be difficult for individuals to meet their nutritional needs within caloric limits. The US FDA requires that nutrition labels list a food's quantity of added sugars [151].

Naturally occurring sugars (eg, from fruit or milk) are not considered added sugars. However, fruit juices should still be limited to small daily amounts, if any, as they tend to be high in calories without the added benefit of fiber, which is found in whole fruit.

Soft drinks and other sugar-sweetened beverages (eg, fruit drinks, sports drinks, and energy drinks) should also be avoided [25,26]. Sugar-sweetened beverages are a key contributor to weight gain and obesity [101]. Their consumption is also associated with lower intake of key nutrients because they replace nutrient-dense foods in the diet. In addition to excess weight, intake of sugar-sweetened beverages is associated with an increased risk of coronary heart disease, type 2 diabetes, hypertension, gout, metabolic syndrome, and all-cause mortality [152-159]. Individuals should be encouraged to drink plain water rather than sweetened beverages.

Artificial sweeteners — Consumption of foods and beverages containing artificial sweeteners generally does not produce long-term weight loss and may be associated with health risks, including increased risks of cardiovascular disease, obesity, diabetes, incident depression, and all-cause mortality. The health effects of nonnutritive sweeteners are discussed separately. (See "[Overview of nonnutritive sweeteners](#)", [section on 'Health effects of NNS consumption'](#).)

Guidelines from the World Health Organization recommend against the use of non-sugar sweeteners (eg, aspartame, sucralose, stevia, saccharin) for weight loss or the reduction of chronic disease risk [160,161].

Red and processed meats — We counsel patients to limit red (eg, beef, lamb, veal, venison, goat) and processed (eg, sausage, hot dogs, beef jerky, bologna) meat, because increased consumption has been associated with a small to moderate increased risk of numerous diseases and premature mortality [162,163]. Although modest, and unconfirmed by randomized trials, these risks are potentially important given the high level of red meat consumption in the United States and increasing consumption of red meat globally. In addition, red and processed meat are typically high in saturated fat, which has been shown to increase cardiovascular risk. (See '[Unhealthy fats](#)' above.)

As examples, in meta-analyses and pooled analyses of prospective cohort studies, increased consumption of red and processed meat has been associated with an increased risk of type 2 diabetes [164], stroke [165], and multiple cancers [166-168], as well as increased cardiovascular disease mortality [169].

The association with negative health effects appears to be stronger and more consistently demonstrated for processed compared with unprocessed red meat [170-175].

In the 2021 Prospective Urban Rural Epidemiology (PURE) Study, an international cohort study including over 134,000 participants followed for 9.5 years, a higher risk of major CVD and mortality was noted among those whose diets contained high amounts of processed meat (HR 1.46, 95% CI 1.08-1.98; and HR 1.51, 95% CI 1.08-2.10, respectively); however, the risks were not increased with consumption of unprocessed red meat [176].

In addition, the reduction of red or processed meat appears to be very modestly associated with reduced mortality. As an example, in a meta-analysis of observational studies, decreased consumption of processed red meat and unprocessed red meat were both associated with a reduction in cardiovascular and all-cause mortality, but the magnitude of reduction was greater and more certain for processed red meat (all-cause mortality: RR 0.92 [95% CI 0.87-0.96] with reduction of processed meat; RR 0.93 [95% CI 0.87-1.00] with reduction of unprocessed meat) [173].

Our guidance to limit or avoid red and processed meats aligns with guidance from the American Heart Association [24,177] and American Cancer Society [23]. In contrast, other guidelines, including the 2025 to 2030 Dietary Guidelines for Americans, do not advise reducing unprocessed red meat [26,37].

Of note, the International Agency for Research on Cancer classifies processed meat as a Group 1 human carcinogen, and red meat as "probably carcinogenic to humans" [178]. (See "[Epidemiology and risk factors for colorectal cancer](#)", [section on 'Red and processed meat'](#).)

Highly processed foods — We advise limiting highly processed foods as much as possible as part of a healthy diet [26]. A growing body of evidence links consumption of highly processed foods with health risks [156,163,179-187]. Although the definition of highly or ultra-processed foods varies, it typically includes items such as sweet and savory snacks, reconstituted meats, packaged frozen entrées, instant soups, and sweetened drinks. Highly processed foods are often calorie-dense; higher in refined grains, sugar, saturated fat, and salt; and lower in fiber and other important nutrients.

Data from meta-analyses of large, observational studies suggest an association between consumption of highly processed foods and a range of adverse health outcomes [188]. As an example, in one systematic review, exposure to ultraprocessed foods was associated with 32 adverse health parameters, including increased risks of all-cause and cardiovascular disease mortality, type 2 diabetes, cancer, cardiovascular events, anxiety, and combined common mental health disorders [189]. The quality of evidence ranged from low to very low for most of these associations.

Alcohol — While certain diets and dietary guidelines allow for moderate alcohol consumption, growing evidence supports the message that zero alcohol is the best choice for patients' overall health [148]. Even light alcohol consumption of three to six drinks per week has been associated with a small increased risk for breast cancer in females as well as increased risk for oropharyngeal and esophageal cancer [190]. A Global Burden of Disease Study analysis found that cancer risk increased monotonically with increasing alcohol intake [148]. Although moderate alcohol use may have beneficial effects on cardiovascular health limited to older adults, the increased risks of cancer and other harms likely offset such benefits [148,191]. (See "[Overview of the risks and benefits of alcohol consumption](#)", [section on 'Cardiovascular disease'](#).)

The risks and benefits of alcohol consumption are discussed in detail elsewhere. (See "[Overview of the risks and benefits of alcohol consumption](#)", [section on 'Is there a safe dose of alcohol?'](#).)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Vitamin deficiencies](#)" and "[Society guideline links: Healthy diet in adults](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see "[Patient education: Healthy eating for adults \(The Basics\)](#)" and "[Patient education: Making healthy choices when eating out \(The Basics\)](#)" and "[Patient education: Vegetarian or vegan diet \(The Basics\)](#)" and "[Patient education: Vitamin supplements \(The Basics\)](#)" and "[Patient education: Can foods or supplements lower cholesterol? \(The Basics\)](#)" and "[Patient education: DASH diet \(The Basics\)](#)" and "[Patient education: Good](#)

food sources of iron (The Basics)" and "Patient education: Low-carbohydrate diet (The Basics)" and "Patient education: Low-fat diet (The Basics)" and "Patient education: Mediterranean diet (The Basics)" and "Patient education: Anti-inflammatory diet (The Basics)"

- Beyond the Basics topics (see "Patient education: Diet and health (Beyond the Basics)" and "Patient education: High-fiber diet (Beyond the Basics)")

SUMMARY AND RECOMMENDATIONS

- **Impact of a healthy diet** – Healthy food choices can extend longevity and reduce the risk of cardiovascular disease, diabetes, hypertension, and cancer, among other important chronic diseases. A healthy diet can also help with weight management and improve health-related quality of life. (See '[Impact of a healthy diet](#)' above.)
- **Choosing a healthy diet or dietary pattern** – The specific diet or dietary pattern an individual chooses is influenced by a wide range of factors, including personal preference, social and cultural influences, as well as economic considerations. Clinicians should work with individuals to determine the best healthy diet pattern based upon their unique circumstances. (See '[Choosing a healthy diet: Patient counseling](#)' above.)
- **What are healthy dietary patterns?** – Specific dietary guidelines may vary, but in general, a high-quality, healthy diet emphasizes limited consumption of red and processed meats, unhealthy fats (saturated and industrial trans fats), sugar, sodium, and alcohol, and a greater intake of fruits, vegetables, legumes, nuts, and whole grains. (See '[Dietary guidelines: Recommended dietary patterns](#)' above.)
 - Examples of healthy dietary patterns associated with health benefits include the Mediterranean diet, the Dietary Approaches to Stop Hypertension (DASH) diet, plant-based and vegetarian diets, a low-fat diet, and a low-cholesterol diet. Most of these dietary patterns share some similar aspects but are further defined by regional characteristics or a specific focus or restriction on dietary components or types of foods ([table 2](#) and [table 3](#) and [table 4](#)). (See '[Mediterranean diet](#)' above and '[DASH diet](#)' above and '[Plant-based and vegetarian diets](#)' above and '[Low-fat diet](#)' above and '[Low-cholesterol diet](#)' above.)
- **Specific healthy dietary components** – Although we focus on overall dietary patterns, individual dietary components can play an important role in nutrition and health outcomes and can offer patients specific areas to focus their efforts at improving their diet. (See '[Dietary components associated with health benefits](#)' above.)
 - Specific healthy dietary components include fruits and vegetables, whole grains, fiber, dairy, and certain protein-rich foods, such as nuts, fish, and soy products. (See '[Fruits and vegetables](#)' above and '[Whole grains](#)' above and '[Fiber](#)' above and '[Reduced-fat dairy](#)' above and '[Protein-rich foods](#)' above.)
- **Dietary patterns with limited evidence of health benefits** – The list of diets with limited evidence of benefit is very long and potentially unlimited, as new diets and eating patterns regularly emerge. Examples of dietary patterns with limited supporting evidence include a low-carbohydrate diet, an organic diet, an anti-inflammatory diet, and a gluten-free diet. (See '[Low-carbohydrate diet](#)' above and '[Organic diet](#)' above and '[Anti-inflammatory diet](#)' above and '[Gluten-free diet](#)' above.)
- **Dietary patterns and specific components to avoid** – Individuals should avoid consumption of a typical "Western" diet (characterized by consumption of red and processed meats, refined grains, high-sodium foods, and sugar, as well as low amounts of fiber, whole grains, legumes, and fruits and vegetables), and avoid or minimize consumption of unhealthy fats (ie, saturated and trans fats), added sugars (including sugar-sweetened beverages), red and processed meats, highly processed foods, and alcohol. (See '[Dietary patterns and components to limit or avoid](#)' above.)

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GRAPHICS

Table 1: Estimated caloric needs per day, by age, sex, and physical activity level

Males				Females [◇]			
Age	Sedentary*	Moderately active [¶]	Active ^Δ	Age	Sedentary*	Moderately active [¶]	Active ^Δ
2	1000	1000	1000	2	1000	1000	1000
3	1000	1400	1400	3	1000	1200	1400
4	1200	1400	1600	4	1200	1400	1400
5	1200	1400	1600	5	1200	1400	1600
6	1400	1600	1800	6	1200	1400	1600
7	1400	1600	1800	7	1200	1600	1800
8	1400	1600	2000	8	1400	1600	1800
9	1600	1800	2000	9	1400	1600	1800
10	1600	1800	2200	10	1400	1800	2000
11	1800	2000	2200	11	1600	1800	2000
12	1800	2200	2400	12	1600	2000	2200
13	2000	2200	2600	13	1600	2000	2200
14	2000	2400	2800	14	1800	2000	2400
15	2200	2600	3000	15	1800	2000	2400
16	2400	2800	3200	16	1800	2000	2400
17	2400	2800	3200	17	1800	2000	2400
18	2400	2800	3200	18	1800	2000	2400
19 to 20	2600	2800	3000	19 to 20	2000	2200	2400
21 to 25	2400	2800	3000	21 to 25	2000	2200	2400
26 to 30	2400	2600	3000	26 to 30	1800	2000	2400
31 to 35	2400	2600	3000	31 to 35	1800	2000	2200
36 to 40	2400	2600	2800	36 to 40	1800	2000	2200
41 to 45	2200	2600	2800	41 to 45	1800	2000	2200
46 to 50	2200	2400	2800	46 to 50	1800	2000	2200
51 to 55	2200	2400	2800	51 to 55	1600	1800	2200
56 to 60	2200	2400	2600	56 to 60	1600	1800	2200
61 to 65	2000	2400	2600	61 to 65	1600	1800	2000
66 to 70	2000	2200	2600	66 to 70	1600	1800	2000
71 to 75	2000	2200	2600	71 to 75	1600	1800	2000
76 and up	2000	2200	2400	76 and up	1600	1800	2000

* Sedentary means a lifestyle that includes only the physical activity of independent living.

¶ Moderately active means a lifestyle that includes physical activity equivalent to walking about 1.5 to 3 miles per day at 3 to 4 miles per hour, in addition to the activities of independent living.

Δ Active means a lifestyle that includes physical activity equivalent to walking more than 3 miles per day at 3 to 4 miles per hour, in addition to the activities of independent living.

◇ Estimates for females do not include women who are pregnant or breastfeeding.

Reproduced from: US Department of Agriculture and U.S. Department of Health and Human Services. Dietary Guidelines for Americans: 2020–2025, 9th ed, December 2020. Available at: <https://www.dietaryguidelines.gov/> (Accessed on June 23, 2022).

Graphic 107834 Version 5.0

Table 2: Healthy Mediterranean-style dietary pattern for children and adults

CALORIE LEVEL OF PATTERN*	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800	3000	3200
FOOD GROUP OR SUBGROUP	Daily amount [†] of food from each group (vegetable and protein foods subgroup amounts are per week)											
Vegetables (cup eq/day)	1	1½	1½	2	2½	2½	3	3	3½	3½	4	4
	Vegetable subgroups in weekly amounts											
Dark-green vegetables (cup eq/wk)	½	1	1	1½	1½	1½	2	2	2½	2½	2½	2½
Red and orange vegetables (cup eq/wk)	2½	3	3	4	5½	5½	6	6	7	7	7½	7½
Beans, peas, lentils (cup eq/wk)	½	½	½	1	1½	1½	2	2	2½	2½	3	3
Starchy vegetables (cup eq/wk)	2	3½	3½	4	5	5	6	6	7	7	8	8
Other vegetables (cup eq/wk)	1½	2½	2½	3½	4	4	5	5	5½	5½	7	7
Fruits (cup eq/day)	1	1	1½	2	2	2½	2½	2½	2½	3	3	3
Grains (ounce eq/day)	3	4	5	5	6	6	7	8	9	10	10	10
Whole grains (ounce eq/day) ^Δ	1½	2	2½	3	3	3	3½	4	4½	5	5	5
Refined grains (ounce eq/day)	1½	2	2½	2	3	3	3½	4	4½	5	5	5
Dairy (cup eq/day) ^Δ	2	2½	2½	2	2	2	2	2½	2½	2½	2½	2½
Protein foods (ounce eq/day)	2	3	4	5½	6	6½	7	7½	7½	8	8	8
	Protein foods subgroups in weekly amounts											
Meats, poultry, eggs (ounce eq/wk)	10	14	19	23	23	26	28	31	31	33	33	33
Seafood (ounce eq/wk) [◇]	3	4	6	11	15	15	16	16	17	17	17	17
Nuts, seeds, soy products (ounce eq/wk)	2	2	3	4	4	5	5	5	5	6	6	6
Oils (grams/day)	15	17	17	22	24	27	29	31	34	36	44	51
Limit on calories for other uses (kcal/day) [§]	130	80	90	120	140	240	250	280	300	330	400	540
Limit on calories for other uses (%/day)	13%	7%	6%	8%	8%	12%	11%	12%	12%	12%	13%	17%

The total dietary pattern should not exceed Dietary Guidelines limits for added sugars, saturated fat, and alcohol; be within the Acceptable Macronutrient Distribution Ranges for protein, carbohydrate, and total fats; and stay within calorie limits. Values are rounded.

* Patterns at 1000, 1200, and 1400 kcal levels are designed to meet the nutritional needs of children ages 2 through 8 years. Patterns from 1600 to 3200 kcal are designed to meet the nutritional needs of children 9 years and older and adults. If a child 4 through 8 years of age needs more energy and, therefore, is following a pattern at 1600 calories or more, their recommended amount from the dairy group should be 2½ cup eq per day. Amount of dairy for children ages 9 through 18 is 3 cup eq per day regardless of calorie level. The 1000 and 1200 kcal level patterns are not intended for children 9 and older or adults. The 1400 kcal level is not intended for children ages 10 and older or adults.

¶ Food group amounts shown in cup equivalents (cup eq) or ounce equivalents (ounce eq). Oils are shown in grams. Quantity equivalents for each food group are: Vegetables, fruits (1 cup eq): 1 cup raw or cooked vegetable or fruit; 1 cup vegetable or fruit juice; 2 cups leafy salad greens; ½ cup dried fruit or vegetable. Grains (1 ounce eq): ½ cup cooked rice, pasta, or cereal; 1 ounce dry pasta or rice; 1 medium (1 ounce) slice bread, tortilla, or flatbread; 1 ounce of ready-to-eat cereal (about 1 cup of flaked cereal). Dairy (1 cup eq): 1 cup milk, yogurt, or fortified soymilk; 1½ ounces natural cheese such as cheddar cheese or 2 ounces of processed cheese. Protein foods (1 ounce eq): 1 ounce lean meats, poultry, or seafood; 1 egg; ¼ cup cooked beans or tofu; 1 tbsp nut or seed butter; ½ ounce nuts or seeds.

Δ Amounts of dairy recommended for children and adolescents are as follows, regardless of the calorie level of the pattern: for age 2 years, 2 cup eq per day; for ages 3 through 8 years, 2½ cup eq per day; for ages 9 through 18 years, 3 cup eq per day.

◇ The US Food and Drug Administration (FDA) and the US Environmental Protection Agency (EPA) provide joint advice regarding seafood consumption to limit methylmercury exposure for women who might become or are pregnant or lactating, and children. Depending on body weight, some women and children should choose seafood lowest in methylmercury or eat less seafood than the amounts in the healthy US-style dietary Pattern. For more information, see the [FDA](#) and [EPA](#) websites.

§ Foods are assumed to be in nutrient-dense forms; lean or low-fat; and prepared with minimal added sugars, refined starches (which are a source of calories but few or no other nutrients), saturated fat, or sodium. If all food choices to meet food group recommendations are in nutrient-dense forms, a small number of calories remain within the overall limit of the pattern (ie, limit on calories for other uses). The amount of calories depends on the total calorie level of the pattern and the amounts of food from each food group required to meet nutritional goals. Calories up to the specified limit can be used for added sugars, saturated fat, and/or alcohol (for nonpregnant adults of legal drinking age only) or to eat more than the recommended amount of food in a food group.

Adapted from: Dietary Guidelines for Americans, 2020-2025. US Department of Agriculture and US Department of Health and Human Services 2020. Available at: <https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials> (Accessed on June 23, 2022).

Table 3: Healthy vegetarian dietary pattern for children and adults

CALORIE LEVEL OF PATTERN*	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800	3000	3200
FOOD GROUP OR SUBGROUP	Daily amount [†] of food from each group (vegetable and protein foods subgroup amounts are per week)											
Vegetables (cup eq/day)	1	1½	1½	2	2½	2½	3	3	3½	3½	4	4
	Vegetable subgroups in weekly amounts											
Dark-green vegetables (cup eq/wk)	½	1	1	1½	1½	1½	2	2	2½	2½	2½	2½
Red and orange vegetables (cup eq/wk)	2½	3	3	4	5½	5½	6	6	7	7	7½	7½
Beans, peas, lentils (cup eq/wk) ^Δ	½	½	½	1	1½	1½	2	2	2½	2½	3	3
Starchy vegetables (cup eq/wk)	2	3½	3½	4	5	5	6	6	7	7	8	8
Other vegetables (cup eq/wk)	1½	2½	2½	3½	4	4	5	5	5½	5½	7	7
Fruits (cup eq/day)	1	1	1½	1½	1½	2	2	2	2	2½	2½	2½
Grains (ounce eq/day)	3	4	5	5½	6½	6½	7½	8½	9½	10½	10½	10½
Whole grains (ounce eq/day)	1½	2	2½	3	3½	3½	4	4½	5	5½	5½	5½
Refined grains (ounce eq/day)	1½	2	2½	2½	3	3	3½	4	4½	5	5	5
Dairy (cup eq/day)	2	2½	2½	3	3	3	3	3	3	3	3	3
Protein foods (ounce eq/day)	1	1½	2	2½	3	3½	3½	4	4½	5	5½	6
	Protein foods subgroups in weekly amounts											
Eggs (ounce eq/wk)	2	3	3	3	3	3	3	3	3	4	4	4
Beans, peas, lentils (ounce eq/wk) ^Δ	1	2	4	4	6	6	6	8	9	10	11	12
Soy products (ounce eq/wk)	2	3	4	6	6	8	8	9	10	11	12	13
Nuts, seeds (ounce eq/wk)	2	2	3	5	6	7	7	8	9	10	12	13
Oils (grams/day)	15	17	17	22	24	27	29	31	34	36	44	51
Limit on calories for other uses (kcal/day) [◇]	170	140	160	150	150	250	290	350	350	350	390	500

Limit on calories for other uses (%/day)	17%	12%	11%	9%	8%	13%	13%	15%	13%	13%	13%	16%
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Intake should not exceed Dietary Guidelines limits for added sugars, saturated fat, and alcohol; and should be within the Acceptable Macronutrient Distribution Ranges for protein, carbohydrate, and total fats; and stay within calorie limits. Values are rounded.

* Patterns at 1000, 1200, and 1400 kcal levels are designed to meet the nutritional needs of children ages 2 through 8 years. Patterns from 1600 to 3200 kcal are designed to meet the nutritional needs of children 9 years and older and adults. If a child 4 through 8 years of age needs more energy and, therefore, is following a pattern at 1600 calories or more, their recommended amount from the dairy group should be 2½ cup eq per day. Amount of dairy for children ages 9 through 18 is 3 cup eq per day regardless of calorie level. The 1000 and 1200 kcal level patterns are not intended for children 9 and older or adults. The 1400 kcal level is not intended for children ages 10 and older or adults.

¶ Food group amounts shown in cup equivalents (cup eq) or ounce equivalents (ounce eq). Oils are shown in grams. Quantity equivalents for each food group are: Vegetables, fruits (1 cup eq): 1 cup raw or cooked vegetable or fruit; 1 cup vegetable or fruit juice; 2 cups leafy salad greens; ½ cup dried fruit or vegetable. Grains (1 ounce eq): ½ cup cooked rice, pasta, or cereal; 1 ounce dry pasta or rice; 1 medium (1 ounce) slice bread, tortilla, or flatbread; 1 ounce of ready-to-eat cereal (about 1 cup of flaked cereal). Dairy (1 cup eq): 1 cup milk, yogurt, or fortified soymilk; 1½ ounces natural cheese such as cheddar cheese or 2 ounces of processed cheese. Protein foods (1 ounce eq): 1 ounce lean meats, poultry, or seafood; 1 egg; ¼ cup cooked beans or tofu; 1 tbsp nut or seed butter; ½ ounce nuts or seeds.

Δ About half of beans, peas, lentils are shown as vegetables, in cup eq, and half as protein foods, in ounce eq. Beans, peas, lentils in the patterns, in cup eq, is the amount in the vegetable group plus the amount in protein foods group (in ounce eq) divided by 4.

◇ Foods are assumed to be in nutrient-dense forms, lean or low-fat and prepared with minimal added saturated fat, added sugars, refined starches, or salt. If all food choices to meet food group recommendations are in nutrient-dense forms, a small number of calories remain within the overall limit of the pattern (ie, limit on calories for other uses). The amount of calories depends on the total calorie level of the pattern and the amounts of food from each food group required to meet nutritional goals. Calories up to the specified limit can be used for added sugars, added refined starches, saturated fat, alcohol, or to eat more than the recommended amount of food in a food group.

Adapted from: *Dietary Guidelines for Americans, 2020-2025*. US Department of Agriculture and US Department of Health and Human Services 2020. Available at: <https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials> (Accessed on June 23, 2022).

Table 4: Healthy US-style dietary pattern for children and adults

CALORIE LEVEL OF PATTERN*	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800	3000	3200
FOOD GROUP OR SUBGROUP	Daily amount [†] of food from each group (vegetable and protein foods subgroup amounts are per week)											
Vegetables (cup eq/day)	1	1½	1½	2	2½	2½	3	3	3½	3½	4	4
	Vegetable subgroups in weekly amounts											
Dark-green vegetables (cup eq/wk)	½	1	1	1½	1½	1½	2	2	2½	2½	2½	2½
Red and orange vegetables (cup eq/wk)	2½	3	3	4	5½	5½	6	6	7	7	7½	7½
Beans, peas, lentils (cup eq/wk)	½	½	½	1	1½	1½	2	2	2½	2½	3	3
Starchy vegetables (cup eq/wk)	2	3½	3½	4	5	5	6	6	7	7	8	8
Other vegetables (cup eq/wk)	1½	2½	2½	3½	4	4	5	5	5½	5½	7	7
Fruits (cup eq/day)	1	1	1½	1½	1½	2	2	2	2	2½	2½	2½
Grains (ounce eq/day)	3	4	5	5	6	6	7	8	9	10	10	10
Whole grains (ounce eq/day) ^Δ	1½	2	2½	3	3	3	3½	4	4½	5	5	5
Refined grains (ounce eq/day)	1½	2	2½	2	3	3	3½	4	4½	5	5	5
Dairy (cup eq/day)	2	2½	2½	3	3	3	3	3	3	3	3	3
Protein foods (ounce eq/day)	2	3	4	5	5	5½	6	6½	6½	7	7	7
	Protein foods subgroups in weekly amounts											
Meats, poultry, eggs (ounce eq/wk)	10	14	19	23	23	26	28	31	31	33	33	33
Seafood (ounce eq/wk) [◇]	2-3 [§]	4	6	8	8	8	9	10	10	10	10	10
Nuts, seeds, soy products (ounce eq/wk)	2	2	3	4	4	5	5	5	5	6	6	6
Oils (grams/day)	15	17	17	22	24	27	29	31	34	36	44	51
Limit on calories for other uses (kcal/day) [¥]	130	80	90	100	140	240	250	320	350	370	440	580
Limit on calories for other uses (%/day)	13%	7%	6%	6%	8%	12%	11%	13%	13%	13%	15%	18%

The total dietary pattern should not exceed Dietary Guidelines limits for added sugars, saturated fat, and alcohol; be within the Acceptable Macronutrient Distribution Ranges for protein, carbohydrate, and total fats; and stay within calorie limits. Values are rounded.

* Patterns at 1000, 1200, and 1400 kcal levels are designed to meet the nutritional needs of children ages 2 through 8 years. Patterns from 1600 to 3200 kcal are designed to meet the nutritional needs of children 9 years and older and adults. If a child 4 through 8 years of age needs more energy and, therefore, is following a pattern at 1600 calories or more, their recommended amount from the dairy group should be 2½ cup eq per day. Amount of dairy for children ages 9 through 18 is 3 cup eq per day regardless of calorie level. The 1000 and 1200 kcal level patterns are not intended for children 9 and older or adults. The 1400 kcal level is not intended for children ages 10 and older or adults.

¶ Food group amounts shown in cup equivalents (cup eq) or ounce equivalents (ounce eq). Oils are shown in grams. Quantity equivalents for each food group are: Vegetables, fruits (1 cup eq): 1 cup raw or cooked vegetable or fruit; 1 cup vegetable or fruit juice; 2 cups leafy salad greens; ½ cup dried fruit or vegetable. Grains (1 ounce eq): ½ cup cooked rice, pasta, or cereal; 1 ounce dry pasta or rice; 1 medium (1 ounce) slice bread, tortilla, or flatbread; 1 ounce of ready-to-eat cereal (about 1 cup of flaked cereal). Dairy (1 cup eq): 1 cup milk, yogurt, or fortified soymilk; 1½ ounces natural cheese such as cheddar cheese or 2 ounces of processed cheese. Protein foods (1 ounce eq): 1 ounce lean meats, poultry, or seafood; 1 egg; ¼ cup cooked beans or tofu; 1 tbsp nut or seed butter; ½ ounce nuts or seeds.

Δ Amounts of whole grains in patterns for children are less than the minimum of 3 ounce-eq in all patterns recommended for adults.

◇ The US Food and Drug Administration (FDA) and the US Environmental Protection Agency (EPA) provide joint advice regarding seafood consumption to limit methylmercury exposure for women who might become or are pregnant or breastfeeding, and children. Depending on body weight, some women and many children should choose seafood lowest in methylmercury or eat less seafood than the amounts in the healthy US-style eating pattern. For more information, see the [FDA](#) and [EPA](#) websites.

§ If consuming up to 2 ounces of seafood per week, children should only be fed cooked varieties from the "Best Choices" list in the FDA/EPA joint "Advice About Eating Fish," available on both websites. If consuming up to 3 ounces of seafood per week, children should only be fed cooked varieties from the "Best Choices" list that contain even lower methylmercury: flatfish (eg, flounder), salmon, tilapia, shrimp, catfish, crab, trout, haddock, oysters, sardines, squid, pollock, anchovies, crawfish, mullet, scallops, whiting, clams, shad, and Atlantic mackerel. If consuming up to 3 ounces of seafood per week, many commonly consumed varieties of seafood should be avoided because they cannot be consumed at 3 ounces per week by children without the potential of exceeding safe methylmercury limits; examples that should not be consumed include: canned light tuna or white (albacore) tuna, cod, perch, black sea bass.

¥ Foods are assumed to be in nutrient-dense forms, lean or low-fat and prepared with minimal added saturated fat, added sugars, refined starches, or salt. If all food choices to meet food group recommendations are in nutrient-dense forms, a small number of calories remain within the overall limit of the pattern (ie, limit on calories for other uses). The amount of calories depends on the total calorie level of the pattern and the amounts of food from each food group required to meet nutritional goals. Calories up to the specified limit can be used for added sugars, added refined starches, saturated fat, alcohol, or to eat more than the recommended amount of food in a food group.

Adapted from: Dietary Guidelines for Americans, 2020-2025. US Department of Agriculture and US Department of Health and Human Services 2020. Available at: <https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials> (Accessed on June 23, 2022).

Table 5: USDA recommended daily or weekly caloric intake of different food groups (USDA recommended average daily food intake)

Calorie level of pattern *	1000	1200	1400	1600	1800	2000
Food group [¶]	Daily amount ^Δ of food from each group (vegetable and protein foods subgroup amounts are per week)					
Vegetables	1 c-eq	1½ c-eq	1½ c-eq	2 c-eq	2½ c-eq	2½ c-eq
Dark-green vegetables (c-eq/week)	½	1	1	1½	1½	1½
Red and orange vegetables (c-eq/week)	2½	3	3	4	5½	5½
Legumes (beans and peas) (c-eq/week)	½	½	½	1	1½	1½
Starchy vegetables (c-eq/week)	2	3½	3½	4	5	5
Other vegetables (c-eq/week)	1½	2½	2½	3½	4	4
Fruits	1 c-eq	1 c-eq	1½ c-eq	1½ c-eq	1½ c-eq	2 c-eq
Grains	3 oz-eq	4 oz-eq	5 oz-eq	5 oz-eq	6 oz-eq	6 oz-eq
Whole grains [◇] (oz-eq/day)	1½	2	2½	3	3	3
Refined grains (oz-eq/day)	1½	2	2½	2	3	3
Dairy	2 c-eq	2½ c-eq	2½ c-eq	3 c-eq	3 c-eq	3 c-eq
Protein foods	2 oz-eq	3 oz-eq	4 oz-eq	5 oz-eq	5 oz-eq	5½ oz-eq
Seafood (oz-eq/week)	3	4	6	8	8	8
Meats, poultry, eggs (oz-eq/week)	10	14	19	23	23	26
Nuts seeds, soy products (oz-eq/week)	2	2	3	4	4	5
Oils	15 g	17 g	17 g	22 g	24 g	27 g
Limit on calories for other uses, calories (% of calories)^{§¶}	150 (15%)	100 (8%)	110 (8%)	130 (8%)	170 (9%)	270 (14%)

USDA: United States Department of Agriculture.

* Food intake patterns at 1000, 1200, and 1400 calories are designed to meet the nutritional needs of 2- to 8-year-old children. Patterns from 1600 to 3200 calories are designed to meet the nutritional needs of children 9 years and older as well as adults. If a child 4 to 8 years of age needs more calories and, therefore, is following a pattern at 1600 calories or more, their recommended amount from the dairy group should be 2.5 cups per day. Children 9 years and older and adults should not use the 1000-, 1200-, or 1400-calorie patterns.

¶ Foods in each group and subgroup are:

■ **Vegetables**

- Dark-green vegetables: All fresh, frozen, and canned dark-green leafy vegetables and broccoli, cooked or raw: for example, broccoli, spinach, romaine, kale, collard, turnip, and mustard greens.
- Red and orange vegetables: All fresh, frozen, and canned red and orange vegetables or juice, cooked or raw: for example, tomatoes, tomato juice, red peppers, carrots, sweet potatoes, winter squash, and pumpkin.
- Legumes (beans and peas): All cooked from dry or canned beans and peas: for example, kidney beans, white beans, black beans, lentils, chickpeas, pinto beans, split peas, and edamame (green soybeans). Does not include green beans or green peas.
- Starchy vegetables: All fresh, frozen, and canned starchy vegetables: for example, white potatoes, corn, green peas, green lima beans, plantains, and cassava.
- Other vegetables: All other fresh, frozen, and canned vegetables, cooked or raw: for example, iceberg lettuce, green beans, onions, cucumbers, cabbage, celery, zucchini, mushrooms, and green peppers.

■ **Fruits**

- All fresh, frozen, canned, and dried fruits and fruit juices: for example, oranges and orange juice, apples and apple juice, bananas, grapes, melons, berries, and raisins.

■ **Grains**

- Whole grains: All whole-grain products and whole grains used as ingredients: for example, whole-wheat bread, whole-grain cereals and crackers, oatmeal, quinoa, popcorn, and brown rice.

- Refined grains: All refined-grain products and refined grains used as ingredients: for example, white breads, refined grain cereals and crackers, pasta, and white rice. Refined grain choices should be enriched.
- **Dairy**
 - All milk, including lactose-free and lactose-reduced products and fortified soy beverages (soymilk), yogurt, frozen yogurt, dairy desserts, and cheeses. Most choices should be fat-free or low-fat. Cream, sour cream, and cream cheese are not included due to their low-calcium content.
- **Protein foods**
 - All seafood, meats, poultry, eggs, soy products, nuts, and seeds. Meats and poultry should be lean or low-fat and nuts should be unsalted. Legumes (beans and peas) can be considered part of this group as well as the vegetable group, but should be counted in one group only.

Δ Food group amounts shown in cup-(c) or ounce-equivalents (oz-eq). Oils are shown in grams (g). Quantity equivalents for each food group are:

- Vegetables and fruits, 1 cup-equivalent is: 1 cup raw or cooked vegetable or fruit, 1 cup vegetable or fruit juice, 2 cups leafy salad greens, ½ cup dried fruit or vegetable.
- Grains, 1 ounce-equivalent is: ½ cup cooked rice, pasta, or cereal; 1 ounce dry pasta or rice; 1 medium (1 ounce) slice bread; 1 ounce of ready-to-eat cereal (about 1 cup of flaked cereal).
- Dairy, 1 cup-equivalent is: 1 cup milk, yogurt, or fortified soymilk; 1½ ounces natural cheese such as cheddar cheese or 2 ounces of processed cheese.
- Protein foods, 1 ounce-equivalent is: 1 ounce lean meat, poultry, or seafood; 1 egg; ¼ cup cooked beans or tofu; 1 Tbsp peanut butter; ½ ounce nuts or seeds.

◇ Amounts of whole grains in the patterns for children are less than the minimum of 3 oz-eq in all patterns recommended for adults.

§ All foods are assumed to be in nutrient-dense forms, lean or low-fat, and prepared without added fats, sugars, refined starches, or salt. If all food choices to meet food group recommendations are in nutrient-dense forms, a small number of calories remain within the overall calorie limit of the pattern (ie, limit on calories for other uses). The number of these calories depends on the overall calorie limit in the pattern and the amounts of food from each food group required to meet nutritional goals. Nutritional goals are higher for the 1200- to 1600-calorie patterns than for the 1000-calorie pattern, so the limit on calories for other uses is lower in the 1200- to 1600-calorie patterns. Calories up to the specified limit can be used for added sugars, added refined starches, solid fats, alcohol, or to eat more than the recommended amount of food in a food group. The overall eating pattern also should not exceed the limits of less than 10% of calories from added sugars and less than 10% of calories from saturated fats. At most calorie levels, amounts that can be accommodated are less than these limits. For adults of legal drinking age who choose to drink alcohol, a limit of up to 1 drink per day for females and up to 2 drinks per day for males within limits on calories for other uses applies; and calories from protein, carbohydrate, and total fats should be within the acceptable macronutrient distribution ranges (AMDRs).

¥ Values are rounded.

Reproduced from: 2015-2020 Dietary Guidelines for Americans. U.S. Department of Health and Human Services and U.S. Department of Agriculture, 8th Edition, December 2015. <https://www.dietaryguidelines.gov/about-dietary-guidelines/previous-editions/2015-dietary-guidelines> (Accessed on April 7, 2021).

Table 6: Daily nutritional intake goals for children and adults

		Age-sex groups												
		M/F 2-3	F 4-8	F 9-13	F 14- 18	F 19- 30	F 31- 50	F 51+	M 4-8	M 9-13	M 14- 18	M 19- 30	M 31- 50	M 51+
Calorie level assessed	Source of goal	1000	1200	1600	1800	2000	1800	1600	1400	1800	2200	2400	2200	2000
Macronutrients														
Protein (% kcal)	AMDR	5-20	10- 30	10- 30	10- 30	10- 35	10- 35	10- 35	10- 30	10- 30	10- 30	10- 35	10- 35	10- 35
Protein (g)	RDA	13	19	34	46	46	46	46	19	34	52	56	56	56
Carbohydrate (% kcal)	AMDR	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65	45- 65
Carbohydrate (g)	RDA	130	130	130	130	130	130	130	130	130	130	130	130	130
Fiber (g)	14 g/ 1000 kcal	14	17	22	25	28	25	22	20	25	31	34	31	28
Added sugars (% kcal)	DGA	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
Total lipid (% kcal)	AMDR	30- 40	25- 35	25- 35	25- 35	20- 35	20- 35	20- 35	25- 35	25- 35	25- 35	20- 35	20- 35	20- 35
Saturated fatty acids (% kcal)	DGA	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10
18:2 Linoleic acid (g)	AI	7	10	10	11	12	12	11	10	12	16	17	17	14
18:3 Linoleic acid (g)	AI	0.7	0.9	1.0	1.1	1.1	1.1	1.1	0.9	1.2	1.6	1.6	1.6	1.6

AMDR: Acceptable Macronutrient Distribution Ranges; RDA: Recommended Dietary Allowance; DGA: Dietary Guidelines for Americans, 2020-2025; AI: adequate intake.

Adapted from: Dietary Guidelines for Americans, 2020-2025. US Department of Agriculture and US Department of Health and Human Services 2020. Available at: <https://www.dietaryguidelines.gov/resources/2020-2025-dietary-guidelines-online-materials> (Accessed on June 23, 2022).

Table 7: Amount of fiber in different foods

Food	Serving	Grams of fiber
Fruits		
Apple (with skin)	1 medium apple	4.4 g
Avocado	1/2 cup	5.0 g
Banana	1 medium banana	3.1 g
Blueberries	1 cup	3.6 g
Oranges	1 orange	3.1 g
Pear (with skin)	1 medium pear	5.5 g
Prunes	1 cup (pitted)	12.4 g
Raspberries	1 cup	8.0 g
Strawberries	1/2 cup	2.0 g
Juices		
Apple (unsweetened, with added ascorbic acid)	1 cup	0.5 g
Grapefruit (white, canned, sweetened)	1 cup	0.2 g
Grape (unsweetened, with added ascorbic acid)	1 cup	0.5 g
Orange (with pulp)	1 cup	0.7 g
Prune	1 cup	2.6 g
Vegetables		
Cooked		
▪ Carrots	1/2 cup (sliced)	2.3 g
▪ Corn	1/2 cup	2.0 g
▪ Green beans	1 cup	4.0 g
▪ Peas	1 cup	8.8 g
▪ Potato (baked, with skin)	1 medium potato	3.8 g
Raw		
▪ Broccoli	1/2 cup	2.6 g
▪ Celery	1 cup	2.0 g
▪ Cucumber (with peel)	1 cucumber	1.5 g
▪ Lettuce	1 cup (shredded)	0.5 g
▪ Spinach	1 cup	0.7 g
▪ Tomato	1 medium tomato	1.5 g
Legumes		
▪ Baked beans (canned, no salt added)	1/2 cup	5.6 g
▪ Black beans (cooked)	1/2 cup	7.5 g
▪ Chickpeas (cooked)	1/2 cup	6.2 g
▪ Kidney beans (canned)	1/2 cup	5.7 g

▪ Lentils (boiled)	1/2 cup	7.8 g
▪ Lima beans (canned)	1/2 cup	6.6 g
▪ Navy beans (cooked)	1/2 cup	9.5 g
Breads, pastas, flours		
Bran muffins	1 medium muffin	5.2 g
Bread (rye)	1 slice	2.0 g
Bread (white)	1 slice	0.6 g
Bread (whole wheat)	1 slice	1.9 g
Bulgur (cooked)	1/2 cup	4.1 g
Cracker (rye wafers)	2 crackers	5.0 g
Oatmeal (cooked)	1 cup	4.0 g
Pasta and rice (cooked)		
▪ Macaroni	1 cup	2.5 g
▪ Rice (brown)	1 cup	3.5 g
▪ Rice (white)	1 cup	0.6 g
▪ Spaghetti (regular)	1 cup	2.5 g
▪ Spaghetti (whole wheat)	1 cup	5.4 g
Pearl barley (cooked)	1 cup	6.0 g
Popcorn (air popped)	3 cups	3.6 g
Breakfast cereals*		
Bran flakes	3/4 cup	5.5 g
"High-fiber" cereals	1/2 cup	14 g
Instant oatmeal	1 pouch (43 g)	3.0 g
Other cereals	1/2 cup	3.0 to 7.5 g
Nuts		
Almonds	1/2 cup	8.7 g
Peanuts	1/2 cup	7.9 g

There are 2 kinds of fiber:

- "Soluble fiber" is found in fruits, oats, barley, beans, and peas. It can help heart health, including lowering cholesterol.
- "Insoluble fiber" is found in wheat, rye, and other grains. It can help with constipation.

The exact amount of fiber in a food depends on the specific product and brand. To learn how much fiber and other nutrients are in different foods, visit the [United States Department of Agriculture FoodData Central website](https://fdc.nal.usda.gov).

* Ready-to-eat breakfast cereals have a wide range of fiber content. Types with high fiber content typically include bran (wheat or oat) and/or oat fiber. Check the nutrition label of each brand for actual fiber content.

Data from: USDA FoodData Central. Available at: <https://fdc.nal.usda.gov> (Accessed on December 6, 2024).

Table 8: Sources and main effects of dietary fat

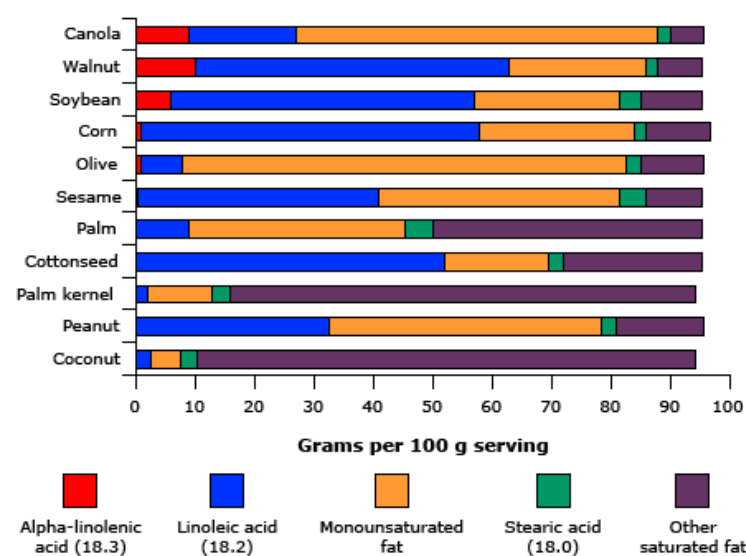
Type of fat	Chief food sources	Leading food contributors in diets of adults in the United States*	Effects on cholesterol	Effects on coronary heart disease
Trans fatty acids, from partially hydrogenated vegetable oils	Stick and full-fat margarine; commercial baked goods; deep-fried foods	Fast food; margarines; commercial baked goods (sweet rolls, cookies, donuts)	Increases LDL cholesterol, lowers HDL cholesterol	Increases risk of coronary heart disease
Saturated fatty acids	Dairy foods; red meat; some plant oils (coconut, palm)	Dairy foods, especially cheese, milk, ice cream; red meat	Increases total cholesterol	May increase risk of coronary heart disease
Monounsaturated fatty acids	Vegetable sources (canola, olive oil); also from meat, dairy	Beef; margarines; chicken; olive oil	Lowers LDL cholesterol and triglycerides, maintains HDL cholesterol	Probably has no association
Polyunsaturated fatty acids; n-6	Safflower, sunflower, and corn oils	Mayonnaise; margarines; salad dressing; nuts; chicken; peanut butter	Lowers LDL cholesterol and triglycerides, increases HDL	May reduce risk of coronary heart disease
Polyunsaturated fatty acids; n-3	Canola, soybean, flaxseed, walnut oil; wheat germ; vegetables of cabbage family For longer-chain n-3 fatty acids: seafood, especially fatty fish	Alpha-linolenic acid (18:3): mayonnaise, salad dressing, margarines, beef; longer-chain n-3: tuna, other dark fish, shrimp	Lowers LDL cholesterol and triglycerides, maintains HDL cholesterol	May reduce risk of coronary heart disease

Leading food contributors combines the fatty acid content of the food with the frequency with which individuals eat that food.

HDL: high-density lipoprotein; LDL: low-density lipoprotein.

* Data from participants in the ongoing Nurses' Health Study (women) and Health Professionals Follow-up Study (men).

Figure 1: Fat content of oils



When fats are needed for cooking, spreads, and other uses, oils with higher amounts of monounsaturated and/or polyunsaturated fatty acids should be used, such as canola oil and olive oil. Those oils that are higher in n-3 polyunsaturated fats (alpha-linolenic acid), such as walnut oil and, to a lesser extent, soybean oil, are also beneficial. Linoleic acid is an n-6 polyunsaturated fatty acid.

Table 9: Foods and products that may contain gluten

Frequently overlooked foods that may contain gluten and need to be verified:	NOT ALLOWED in any form:
▪ Brown rice syrup ▪ Breading and coating mixes ▪ Croutons ▪ Energy bars ▪ Flour or cereal products ▪ Imitation bacon ▪ Imitation seafood ▪ Marinades ▪ Panko (Japanese bread crumbs) ▪ Pastas ▪ Processed luncheon meats ▪ Sauces, gravies ▪ Self-basting poultry ▪ Soy sauce or soy sauce solids ▪ Soup bases ▪ Stuffings, dressing ▪ Thickeners (roux) ▪ Communion wafers ▪ Herbal supplements ▪ Probiotic products* ▪ Drugs and over-the-counter medications ▪ Nutritional supplements ▪ Vitamins and mineral supplements ▪ Play-Doh, crayons, paint, glue, paper mache – A potential problem if the child puts their hands on or in the mouth while playing; wash hands after using these products	▪ Wheat (einkorn, durum, faro, graham, kamut, semolina, spelt) ▪ Rye ▪ Barley ▪ Triticale ▪ Malt, malt flavoring, malt vinegar (are generally made from barley; verify the source)

* In 2015, a study of 22 commonly used probiotics revealed that 12 (55%) contained gluten, including 2 that were labeled gluten-free despite containing gluten levels higher than the 20 parts per million required for gluten-free labeling.^[1]

Reference:

1. Nazareth S, Lebwohl B, Voyksner JS, Green PH. Widespread Contamination of Probiotics With Gluten, Detected by Liquid Chromatography-Mass Spectrometry. *Gastroenterology* 2015; 148:S28.

Modified with permission: "Frequently overlooked foods that may contain gluten and need to be verified." From *Quick Start Diet Guide for Celiac Disease* © 2009. Celiac Disease Foundation. www.celiac.org.

Contributor Disclosures

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